

HALKIRK AGCM, AB, Canada

WMO: 717920

Lat: 52.1166N

Lon: 112.1641W

Elev: 843

StdP: 91.60

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-30.5	-27.9	-33.9	0.2	-30.4	-31.3	0.2	-28.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.6	29.2	16.4	27.2	15.9	25.3	15.3	18.5	24.9	17.4	24.5	16.4	23.2	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.2	12.8	21.4	15.0	11.8	19.9	13.8	10.9	18.7	56.0	24.7	52.2	24.5	49.2	23.7	23.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	-35.2	32.4	2.5	2.6	-37.0	34.3	-38.4	35.8	-39.8	37.3	-41.6	39.2
			-34.6	20.7	3.2	1.5	-36.9	21.8	-38.8	22.7	-40.5	23.6	-42.8	24.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.8	-10.0	-11.3	-4.7	2.8	10.1	14.1	16.7	15.9	10.9	3.8	-4.5	-11.3
	DBStd	11.95	7.95	8.30	7.91	5.42	4.37	3.09	2.84	3.32	4.58	5.09	7.25	7.83
	HDD10.0	3293	619	597	457	221	52	5	0	2	43	201	437	659
	HDD18.3	5708	877	831	716	467	254	130	66	89	223	452	686	918
	CDD10.0	662	0	0	0	4	57	128	208	184	71	8	1	0
	CDD18.3	35	0	0	0	0	1	3	16	14	2	0	0	0
	CDH23.3	884	0	0	0	3	52	86	319	318	103	3	0	0
	CDH26.7	223	0	0	0	0	6	13	88	91	25	0	0	0
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	PrecAvg	423	21	12	18	31	47	79	77	57	34	19	15	14
(15) Precipitation	PrecMax	529	63	26	38	60	85	114	166	107	77	45	56	30
	PrecMin	289	1	3	2	4	16	25	24	6	3	5	3	5
	PrecStd	71	15	7	9	16	21	26	34	24	20	9	12	7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.4	6.4	13.8	21.7	27.3	28.4	31.5	31.7	29.7	22.7	14.8	6.9
		MCWB	2.9	2.9	7.2	9.7	13.1	16.0	18.0	17.0	15.0	12.2	7.4	3.2
	2%	DB	4.7	3.5	10.1	18.3	24.6	25.7	29.2	29.0	26.2	18.1	10.6	3.9
		MCWB	1.9	0.7	4.9	8.7	12.0	15.2	17.2	16.3	13.9	9.8	5.4	0.9
	5%	DB	3.2	1.7	7.1	15.0	22.2	23.7	26.9	26.9	23.5	15.1	6.8	1.5
		MCWB	0.9	-0.4	3.3	6.6	11.3	14.5	17.0	16.1	13.2	8.1	3.0	-0.8
	10%	DB	1.0	-0.2	4.5	11.9	19.5	21.6	24.7	24.6	20.4	12.0	4.4	-0.7
		MCWB	-1.0	-1.6	1.6	5.1	10.2	13.6	16.4	15.6	12.0	6.6	1.4	-2.4
	0.4%	WB	3.1	3.1	7.8	10.9	14.2	18.3	20.9	19.6	16.3	12.8	8.0	3.6
		MCDB	5.8	6.0	13.4	20.6	24.5	24.8	27.3	25.5	26.1	21.7	14.4	6.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	2.1	1.0	5.1	8.8	12.9	16.5	18.9	18.3	14.8	10.5	5.5	1.1
		MCDB	4.4	3.1	9.5	17.4	21.8	23.2	25.0	24.8	24.3	16.2	10.0	3.8
	5%	WB	1.0	-0.2	3.4	7.1	12.0	15.4	17.7	17.2	13.5	8.7	3.3	-0.6
		MCDB	3.2	1.5	6.9	14.0	20.7	21.4	24.6	24.8	21.8	14.2	6.2	1.3
	10%	WB	-0.9	-1.6	1.7	5.6	10.9	14.3	16.9	16.1	12.4	7.0	1.7	-2.3
		MCDB	1.0	-0.1	4.1	11.2	18.0	20.2	23.7	22.9	20.1	11.6	3.9	-0.7
(35) Mean Daily Temperature Range	5% DB	MDBR	9.7	10.0	10.2	11.9	13.4	12.2	13.6	14.0	13.7	12.0	9.8	9.3
		MCDBR	9.9	10.6	12.8	18.0	17.7	15.9	18.1	18.9	19.6	16.7	12.8	10.6
		MCWBR	7.8	8.6	8.2	9.3	7.8	7.1	7.9	8.2	9.0	9.3	8.2	8.5
	5% WB	MCDBR	10.3	10.4	12.2	16.4	15.4	13.9	15.7	15.8	17.9	15.7	11.9	10.7
		MCWBR	8.3	8.7	8.1	8.9	7.1	7.0	7.8	7.7	8.6	9.0	8.0	8.8
(40) Clear-Sky Solar Irradiance	taub		0.267	0.283	0.285	0.328	0.353	0.358	0.352	0.366	0.315	0.283	0.256	0.246
	taud		2.293	2.336	2.451	2.367	2.343	2.367	2.418	2.358	2.467	2.537	2.471	2.339
	Ebn at Noon		742	842	921	910	898	894	859	874	845	774	715	
	Edh at Noon		62	80	88	108	117	115	108	109	86	65	52	50
	RadAvg		1.10	2.15	3.55	4.75	5.77	6.07	6.42	5.23	3.67	2.20	1.14	0.83
(45) All-Sky Solar Radiation	RadStd		0.10	0.18	0.27	0.42	0.38	0.39	0.32	0.38	0.38	0.27	0.11	0.08

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (80 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page